



ARTIFICIAL INTELLIGENCE

Class XI

STUDENT HANDBOOK



Subject Code:843

Unit 8: AI Ethics and Values

Title: AI Ethics and Values	Approach: Team discussion, Web search, Case studies
Summary: In the present age of extensive use of AI tools by people of different walks of life, this chapter will guide the students to keep the guidelines in mind during the development and usage of AI. Also, students will be able to understand the ethical implications of different AI tools which have created a lot of commotion in the community because of different types of bias. They will understand the present-day challenges related to AI ethics.	
Learning Objectives: 1. Understand the fundamental concepts of ethics and its relevance in the context of artificial intelligence (AI). 2. Identify bias arising from various sources present in AI systems and understand their societal implications. 3. Understand the importance of mitigating bias in AI systems and be able to identify strategies for reducing bias in AI technologies. 4. Understand the importance of developing AI policies.	
Key Concepts: 1. Ethics in Artificial Intelligence 2. The five pillars of AI Ethics 3. Bias, Bias Awareness, Sources of Bias 4. Mitigating Bias in AI Systems 5. Developing AI Policies 6. Moral Machine Game 7. Survival of the Best Fit Game	
Learning Outcomes: Students will be able to - 1. Demonstrate an understanding of the fundamental principles of ethics and gain insight into ethical considerations related to AI technologies. 2. Develop an understanding of AI bias, its sources, and its real-world implications, as well as the ethical considerations. 3. Identify and apply strategies for mitigating bias in AI systems to promote fairness and transparency in technology. 4. Recognize the significance of AI policies in promoting responsible, safe, and ethical use of AI technologies.	
Pre-requisites: Understanding the concept of data and reasonable fluency in English language. Ability to understand visual data.	

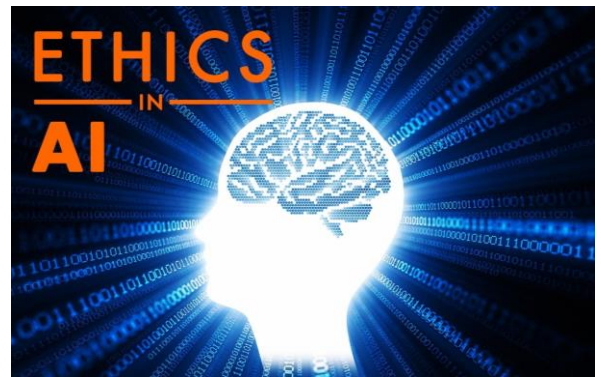
ETHICS IN ARTIFICIAL INTELLIGENCE



In today's rapidly evolving technological landscape, artificial intelligence (AI) has emerged as a transformative force with the potential to revolutionize various aspects of human society. However, with this technological advancement comes a myriad of ethical considerations and challenges. This chapter serves as a foundational exploration into the ethical dimensions of AI, aiming to equip students with the necessary knowledge and

tools to navigate the complex ethical landscape of AI technologies.

Ethics refers to the moral principles that govern human behavior and decision-making. It encompasses concepts such as right and wrong, fairness, justice, and accountability. Ethical considerations guide individuals and organizations in making responsible choices that align with societal values and norms. In the context of AI, ethics play a crucial role in ensuring that AI technologies are developed and deployed in a manner that upholds ethical principles and respects human rights. Ethical considerations are essential in addressing issues such as bias, transparency, accountability, privacy, and the societal impact of AI.



AI ethics refers to the ethical principles and guidelines that govern the design, development, and deployment of AI technologies. AI ethics aims to ensure that AI systems are developed and used in ways that are fair, transparent, accountable, and aligned with human values.

Example 1:

Suppose a CCTV camera was to spot your face in a crowd outside a sports stadium. In the police data center somewhere in the city/ country, an artificial neural network analyzes images from the CCTV footage frame by-frame. A floating cloud in the sky causes a shadow on your face and the neural network (by mistake) finds your face similar to the face of a wanted criminal.

- If the police were to call you aside for questioning and tell you they had reason to detain you, how would you defend yourself?
- Was it your fault that your shadowed face has resemblance by few degrees with a person in the police record?

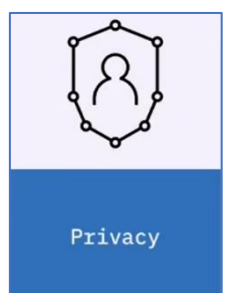
Example 2:

This happened in the USA in 2018. An AI system was being used to allocate care to nearly 200 million patients in the US. It was discovered later that the AI system was offering a lower standard of care to the black patients. Across the board, black people were assigned lower risk scores than white people. This in turn meant that black patients were less likely to be able to access the necessary standard of care.

The problem stemmed from the fact that the AI algorithm was allocating risk values using the predicted cost of healthcare. Because black patients were often less able to pay or were perceived as less able to pay for the higher standard of care, the AI essentially learned that they were not entitled to such a standard of treatment. Though the system was fixed / improved after being discovered, the big question is – whose problem was this? The AI system developers or the US black people data (which was true to an extent)?

THE FIVE PILLARS OF AI ETHICS

- **Explainability** refers to the transparency and interpretability of AI systems, allowing users to understand how algorithms make decisions and predictions. Explainable AI enables stakeholders to comprehend the underlying logic, factors, and considerations driving algorithmic outcomes, fostering trust, accountability, and ethical use of AI technologies. Explainability is essential for ensuring that AI systems are transparent, accountable, and aligned with ethical principles.
- **Fairness** in AI seeks to remove bias and discrimination from algorithms and decision-making models. Machine learning fairness addresses and eliminates algorithmic bias from machine learning models based on sensitive attributes like race and ethnicity, gender, sexual orientation, disability, and socioeconomic class.
- **Robustness** in AI systems indeed refers to their ability to consistently provide accurate and reliable results regardless of the conditions they encounter and for extended periods. It is all about making sure that AI algorithms and systems operate as expected without any unexpected errors or deviations from their intended behavior. This involves ensuring stability in the algorithms, being able to reproduce results, and maintaining consistent performance across different datasets and environments. Achieving reliability in AI systems requires thorough testing, validation, and quality assurance at every stage of development.
- **Transparency** involves openness and disclosure about the design, operation, and implications of AI systems. Transparent AI frameworks provide clear documentation, disclosure, and communication about the data, algorithms, and decision-making processes used in AI applications. Transparency promotes accountability, scrutiny, and informed decision-making, enabling users and stakeholders to assess the ethical implications and societal impacts of AI technologies.
- **Privacy** refers to the right of individuals to control their personal information and to be free from unwanted intrusion into their lives. It encompasses the ability to keep certain aspects of one's life private, such as personal communications, activities, and personal data. Privacy is essential for safeguarding personal autonomy, dignity, and freedom from unwarranted interference.



Activity:

Organize the class into groups. Reflect on the following points based on the video links given

- Video: [AI for Good](#)
- Reflect on the video "[The Ethical Robot](#)" and identify two ethical dilemmas that stood out to you. Document these questions.
- Using "[How to build a moral robot](#)" as a reference point, list the moral and ethical principles you wish to embed in your robot. Consider the video as inspiration, but do not feel constrained by its content. Feel free to expand your thoughts with creativity and innovation.
- Assemble a team of five students and collectively watch the video "[Humans need not apply](#)." It is recommended to view the video multiple times. Following your discussions, compile a group paper summarizing your insights and interpretations from the video.

BIAS, BIAS AWARENESS, SOURCES OF BIAS

Bias, in simple terms, means having a preference or tendency towards something or someone over others, often without considering all the relevant information fairly. It can lead to unfair treatment or decisions based on factors like personal beliefs, past experiences, or stereotypes. In everyday life, bias can affect how we perceive and interact with people, situations, or ideas. In the context of artificial intelligence, bias refers to when AI systems make unfair or inaccurate decisions due to flawed data or built-in assumptions, which can result in unfair outcomes for certain groups of people.

In today's interconnected world, artificial intelligence (AI) technologies play an increasingly prominent role in various aspects of our lives, from healthcare to finance to criminal justice. However, as AI systems become more pervasive, it is essential to recognize and address the presence of bias in these technologies. **Bias awareness** means understanding that AI systems might have unfair preferences because of different things like the information they were taught with, the rules they follow, or the ideas they were built upon. So, being aware of bias in AI is like knowing that sometimes AI might make unfair choices or judgments because of how it was trained or made.

Activity:

Question 1: Why are most images that show up when you do an image search for “vacation” seen as beaches?

Question 2: Why are most images that show up when you do an image search for “nurse” seen as females?

Question 3: Organize students into groups and ask them to find answers for the questions given below after going through the link [Amazon Recruitment Tool](#)

Discussion Questions:

- How do algorithmic hiring systems function, and what criteria are typically used to evaluate job applicants?
- What are the ethical implications of using biased algorithms in hiring processes, particularly regarding fairness, equal opportunity, and diversity?
- How might biased hiring algorithms perpetuate systemic inequalities in employment and hinder efforts to promote inclusivity in the workforce?

Reference: <https://www.ibm.com/blog/shedding-light-on-ai-bias-with-real-world-examples/>

AI bias, also referred to as machine learning bias or algorithm bias, refers to AI systems that produce biased results that reflect and perpetuate human biases within a society, including historical and current social inequality. Bias can be found in the initial training data, the algorithm, or the predictions the algorithm produces. When bias goes unaddressed, it hinders people's ability to participate in the economy and society. It also reduces AI's potential.

The source of bias in AI

Eliminating AI bias requires drilling down into datasets, machine learning algorithms and other elements of AI systems to identify sources of potential bias.

1. Training data bias

AI systems learn to make decisions based on training data, so it is essential to assess datasets for the presence of bias. One method is to review data sampling for over- or underrepresented groups within the training data. For example, training data for a facial recognition algorithm that over-represents white people may create errors when attempting facial recognition for people of color. Similarly, security data that includes information gathered in geographic areas that are predominantly black could create racial bias in AI tools used by police. Bias can also result from how the training data is labeled. For example, AI recruiting tools that use inconsistent labeling or exclude or over-represent certain characteristics could eliminate qualified job applicants from consideration.

2. Algorithmic bias

Using flawed training data can result in algorithms that repeatedly produce errors, unfair outcomes, or even amplify the bias inherent in the flawed data. Algorithmic bias can also be caused by programming errors, such as a developer unfairly weighting factors in algorithm decision-making based on their own conscious or unconscious biases. For example, indicators like income or vocabulary might be used by the algorithm to unintentionally discriminate against people of a certain race or gender.

3. Cognitive bias

When people process information and make judgments, we are inevitably influenced by our experiences and our preferences. As a result, people may build these biases into AI systems through the selection of data or how the data is weighted. For example, cognitive bias could lead to favoring datasets gathered from Americans rather than sampling from a range of populations around the globe.

Examples of AI bias in real life

- **Healthcare**—Underrepresented data of women or minority groups can skew predictive AI algorithms. For example, computer-aided diagnosis (CAD) systems have been found to return lower accuracy results for black patients than white patients.
- **Online advertising**—Biases in search engine ad algorithms can reinforce job role gender bias. Independent research at Carnegie Mellon University revealed that [Google's online advertising system](#) displayed high-paying positions to males more often than to women.
- **Image generation**—[Academic research](#) found bias in the generative AI art generation application Midjourney. When asked to create images of people in specialized professions, it showed both younger and older people, but the older people were always men, reinforcing gender bias of the role of women in the workplace.

Activity: Role Play

Share the following examples of biased AI systems and their potential consequences and ask students to do a role play to present each scenario:

- **Facial Recognition Technology:**

- **Example:** Facial recognition systems have been shown to exhibit bias against certain demographic groups, particularly people with darker skin tones and women.
- **Consequences:** Biased facial recognition algorithms can lead to misidentification and wrongful arrests, disproportionately affecting marginalized communities and eroding trust in law enforcement.

- **Predictive Policing Algorithms:**

- **Example:** Predictive policing algorithms use historical crime data to forecast future criminal activity and allocate law enforcement resources. However, studies have found that these algorithms can perpetuate racial and socioeconomic biases, leading to over-policing of minority neighborhoods.
- **Consequences:** Biased predictive policing algorithms may exacerbate racial profiling and discrimination, fueling tensions between law enforcement agencies and communities of color and undermining public trust in the criminal justice system.

- **Algorithmic Hiring Systems:**

- **Example:** AI-powered hiring systems are used by companies to screen job applications and identify potential candidates. However, research has shown that these systems can perpetuate gender and racial biases, favoring certain demographic groups over others.
- **Consequences:** Biased hiring algorithms may reinforce existing disparities in employment opportunities, leading to discrimination against underrepresented groups and hindering efforts to promote diversity and inclusion in the workforce.

- **Healthcare Algorithms:**

- **Example:** AI algorithms are increasingly used in healthcare for tasks such as diagnosing diseases and predicting patient outcomes. However, studies have identified biases in healthcare algorithms that can result in differential treatment recommendations based on factors such as race or socioeconomic status.
- **Consequences:** Biased healthcare algorithms may lead to disparities in patient care, with certain demographic groups receiving suboptimal or inequitable treatment. This can contribute to worsened health outcomes and perpetuate healthcare inequalities.

- **Credit Scoring Systems:**

- **Example:** AI-powered credit scoring systems are used by financial institutions to assess individuals' creditworthiness and determine their eligibility for loans and other financial products. However, these systems have been found to exhibit biases that disproportionately disadvantage certain demographic groups, such as low-income individuals and people of color.
- **Consequences:** Biased credit scoring algorithms may limit access to financial opportunities for marginalized communities, perpetuating socioeconomic inequalities and hindering economic mobility.

MITIGATING BIAS IN AI SYSTEMS

Mitigating bias in AI systems is essential for several reasons. Firstly, when AI systems have bias, they can make existing problems like unfairness and discrimination even worse. For example, biased algorithms used in hiring processes may unfairly disadvantage certain groups, leading to systemic discrimination. Secondly, biased AI makes people trust technology less. If people don't trust AI to make fair decisions, they might not want to use it, which can cause problems for everyone. Lastly, addressing bias is essential for upholding ethical principles and ensuring that AI technologies are developed and used responsibly.

Strategies for Mitigating Bias

There are several strategies and techniques for mitigating bias in AI systems:

- **Using Diverse Data:** To reduce bias, we should use lots of different kinds of information to teach AI. This way, the AI can learn from many different examples and viewpoints, making it less likely to be biased.
- **Detecting Bias:** We need ways to find and measure bias in AI systems before they are used. This could mean looking at how the AI makes decisions for different groups of people or using special tools to see if the AI is being fair.
- **Fair Algorithms:** We can make AI systems fairer by using special algorithms that are designed to be fair. These algorithms make sure to consider fairness when making decisions, helping to reduce bias.
- **Being Transparent:** It is important for AI systems to be clear and explain how they make decisions. When people understand how AI works, they can see if there is any bias and fix it.
- **Inclusive Teams:** When creating AI, it is helpful to have a team of people from different backgrounds and experiences. This way, they can spot biases that others might miss and make sure the AI is fair for everyone.

Activity:

Allow students to examine various forms of media, such as news articles, advertisements, or social media posts, and identify instances of bias based on factors like race, gender, or socio-economic status. Encourage them to discuss how bias can influence perceptions and stereotypes.

Do you know?

IBM's AI Fairness 360 is an open-source and comprehensive toolkit that offers a wide range of tools and resources for addressing bias in machine learning models. The toolkit includes over 70 fairness metrics which help users detect bias in their models. This suggests that AI Fairness 360 provides robust mechanisms for identifying potential sources of bias. The toolkit offers over 10 algorithms for mitigating bias once it has been detected. These bias mitigating algorithms include optimizing the preprocessing stage, prejudice remover, and regular algorithms, among others. With its diverse set of features, educational resources, and validation mechanisms, it aims to support users in promoting fairness and equity in AI applications.

DEVELOPING AI POLICIES

Developing AI policies is essential for ensuring that AI technologies are used responsibly, safely, and ethically, while also promoting innovation and public trust.

- Rules for AI should start with being good to people and respecting their rights. This means treating everyone fairly, being honest about how AI works, making sure it is safe, and being accountable if something goes wrong.
- We need clear rules and standards for how AI is used. These rules should cover important things like protecting people's information, making sure AI does not have unfair biases, keeping it safe, and making sure people can ask questions about how AI works.
- When making these rules, it is important to talk to lots of different people. This includes government people, business leaders, scientists, community groups, and regular people. Everyone's opinion matters because AI affects everyone.
- Before using AI, we should check to see if there are any problems or risks. This means thinking about what could go wrong and making plans to fix it.

Understanding the components of AI policies involves examining guidelines and principles established by various organizations and regulatory bodies.

1. IBM AI Ethics Board:

Focus: Ethical development and deployment of AI technologies across various industries.

Components:

- Development of ethical principles and guidelines for AI research and development.
- Recommendations for addressing ethical considerations such as fairness, transparency, accountability, and bias mitigation in AI systems.
- Engagement with stakeholders, including researchers, policymakers, and industry partners, to promote dialogue and collaboration on ethical AI practices.
- Support for educational initiatives and resources to raise awareness and understanding of AI ethics among developers, users, and the public.

2. Microsoft's Responsible AI Page:

Focus: Corporate responsibility and ethics in AI

Components:

- Principles for responsible AI development and deployment, including fairness, reliability, privacy, and inclusivity.
- Tools and resources for integrating ethical considerations into AI projects, such as fairness assessments and bias detection algorithms.
- Case studies and best practices for implementing responsible AI practices across various industries and domains.

3. Artificial Intelligence at Google:

Focus: Corporate AI ethics and governance

Components:

- Google's principles for ethical AI development, encompassing areas such as fairness, safety, privacy, and accountability.
- Guidelines for designing AI systems that prioritize human values and societal well-being.

- Commitments to transparency, collaboration, and continuous improvement in AI governance and decision-making.

4. **European Union's Ethics Guidelines for Trustworthy AI—Press Release:**

Focus: Ethical guidelines for AI development and deployment in the EU

Components:

- Principles for trustworthy AI, including respect for human autonomy, prevention of harm, fairness, and accountability.
- Requirements for transparency, explainability, and auditability in AI systems.
- Recommendations for ensuring human oversight and accountability mechanisms in AI applications with high societal impact.

Further reading:

- [IBM's AI Ethics Board](#)
- [Microsoft's Responsible AI page](#)
- [Artificial Intelligence at Google: Our Principles](#)
- [European Union's Ethics Guidelines for Trustworthy AI—Press Release](#)

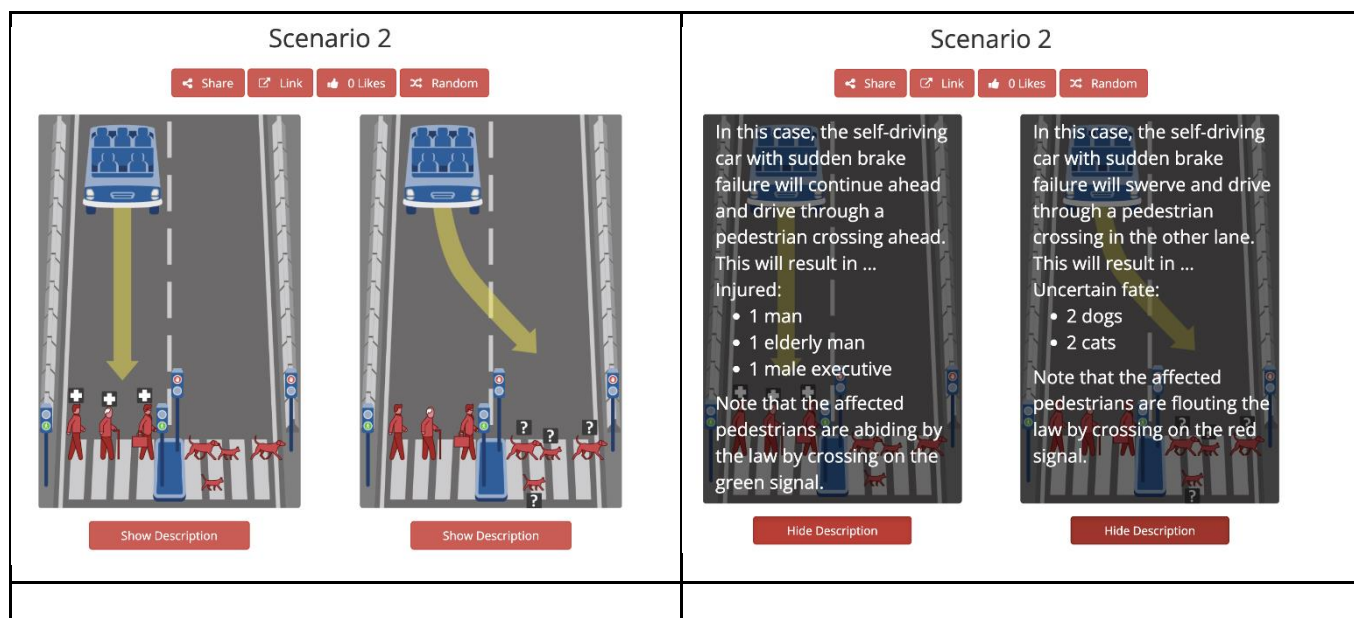
MORAL MACHINE GAME

An **ethical dilemma** is a situation in which a person or group is faced with conflicting moral principles or choices. In other words, it is a scenario where there is no clear "right" or "wrong" decision, and any action taken may have both positive and negative consequences. Ethical dilemmas often involve difficult choices between two or more options, each of which may involve values or principles that are important to the decision-maker.

An **ethical dilemma in the context of artificial intelligence (AI)** arises when there is a conflict between moral principles or values in the design, development, deployment, or use of AI technologies. These dilemmas often emerge due to the complex and multifaceted nature of AI systems, which can have far-reaching implications for individuals, society, and the environment.

The **Moral Machine**, developed by researchers at the Massachusetts Institute of Technology (MIT), is an online platform designed to explore ethical dilemmas in AI through interactive decision-making scenarios. Users are presented with various hypothetical situations where autonomous vehicles must make split-second decisions that could result in harm or even loss of life. The scenarios often involve difficult choices, such as deciding whether to prioritize the safety of passengers or pedestrians, obey traffic laws, or consider factors like age, gender, and social status.

The Moral Machine presents users with a series of thought-provoking scenarios, each with its own set of moral considerations and trade-offs. For example, imagine you are the operator of a self-driving car that encounters a situation where it must choose between swerving to avoid hitting a group of pedestrians, potentially endangering its passengers, or staying the course and risking harm to those on the road. What decision would you make? And more importantly, why?



While the scenarios presented on the Moral Machine are hypothetical, they reflect real-world ethical dilemmas that AI developers, policymakers, and society at large must grapple with. As AI continues to advance and become more pervasive, it is essential that we engage in meaningful dialogue and reflection on the ethical implications of these technologies. The Moral Machine serves as a powerful tool for sparking conversation, raising awareness, and promoting ethical awareness in the age of AI.

Activity:

1. Begin with: [The Ethical Challenges of Autonomous Vehicles](#)

Discussion Points:

- What ethical issues are highlighted by the presenter concerning the application of AI in military contexts, and how do these resonate with your viewpoints?
 - What steps can legislators take to tackle the ethical dilemmas presented by autonomous weaponry, particularly regarding accountability and the preservation of human dignity?
2. Present the Moral Machine game as a tool for delving into ethical decision-making challenges related to AI - accessible at <https://www.moralmachine.net/>

SURVIVAL OF THE BEST FIT GAME

Activity:

Survival of the Best Fit is an educational game about hiring bias in AI. We aim to explain how the misuse of AI can make machines inherit human biases and further inequality.

Students can do this activity in the practical class to understand the concept of bias while people are hired.

Present this tool for deeper understanding of bias during hiring of people in any company or organization- <https://www.survivalofthebestfit.com/>

EXERCISES

A. Multiple Choice Questions

1. What is the primary focus of AI ethics?
 - a. Ensuring AI systems are efficient and error-free
 - b. Guiding the development and use of AI technologies according to ethical principles
 - c. Maximizing the profitability of AI technologies
 - d. Focusing solely on the technological advancements in AI
2. Which ethical principle is concerned with fairness and justice in AI systems?
 - a. Transparency
 - b. Accountability
 - c. Bias and fairness
 - d. Privacy
3. What role does transparency play in AI ethics?
 - a. Ensuring AI systems are invisible to users
 - b. Making the AI decision-making process clear and understandable
 - c. Keeping the AI technologies a secret
 - d. None of the above
4. What is a major ethical concern related to AI and privacy?
 - a. AI systems improving data security automatically
 - b. AI technologies accessing and using personal data without consent
 - c. AI enhancing user privacy by default
 - d. None of the above
5. How can bias in AI systems impact society?
 - a. By making AI systems more efficient
 - b. By perpetuating and exacerbating existing inequalities
 - c. By improving fairness and justice automatically
 - d. None of the above
6. Which of the following strategies is suggested for mitigating bias in AI systems?
 - a. Ignoring the bias
 - b. Increasing the complexity of AI algorithms
 - c. Regular audits and inclusive data practices
 - d. Focusing solely on the technological aspects
7. What is the purpose of ethical frameworks and guidelines in AI?
 - a. To restrict the development of AI technologies
 - b. To ensure AI development aligns with human values and ethical principles
 - c. To make AI systems less transparent
 - d. None of the above
8. Who is responsible for ensuring the ethical use of AI systems?
 - a. AI developers only
 - b. Government bodies only
 - c. Users only
 - d. All stakeholders, including developers, governments, and users

9. What is an example of AI assisting humanity as mentioned in the unit?
 - a. Reducing energy efficiency in buildings
 - b. Enhancing mental health care through AI-powered chatbots
 - c. Increasing bias in hiring processes
 - d. Decreasing crop yield predictions accuracy
10. What is the role of AI ethics in the context of autonomous vehicles?
 - a. To ensure vehicles are aesthetically pleasing
 - b. To navigate ethical dilemmas such as decision-making in critical situations
 - c. To make autonomous vehicles less safe
 - d. None of the above

B. True/False

1. Ethical considerations in AI are only relevant for AI researchers and developers.
2. Bias in AI systems can lead to unfair and discriminatory outcomes.
3. Transparency in AI systems means making the algorithm's code publicly available.
4. Bias in AI systems can lead to unfair outcomes and reinforce existing societal inequalities.
5. Transparency is not important in AI decision-making processes as long as the outcomes are beneficial.
6. Understanding the fundamental concepts of ethics is irrelevant in the context of artificial intelligence.
7. Analyzing real-world examples of AI bias provides insights into the ethical challenges in AI development.
8. Identifying bias in AI systems has no impact on society.
9. Evaluating strategies to mitigate bias in AI systems is unnecessary.
10. The ability to critically analyze the ethical implications of AI technologies does not require considering their impact on society.

C. Fill in the Blanks

1. Understanding the fundamental principles of ethics is crucial to applying ethical considerations in the field of _____.
2. The ability to critically analyze the ethical implications of AI decision-making processes requires a deep understanding of their _____ on individuals and society.
3. Investigating various types of bias in AI systems enables students to understand their _____ implications.
4. _____ in AI systems can lead to unfair and discriminatory outcomes, making it essential to address issues of bias, fairness, and equity.
5. In the context of AI, _____ is important for making the decision-making processes of AI systems clear and understandable to users.

D. Short Answer Type Questions

1. What is algorithmic bias, and can you give an example?
2. How can bias in AI systems be reduced?
3. Define AI ethics
4. Why is transparency important in AI systems?
5. How can AI impact privacy and data protection?
6. What role does fairness play in AI systems?

7. How can ethical dilemmas in AI be addressed?
8. What strategies can mitigate bias in AI?
9. What challenges do policymakers face in regulating AI?

E. Case Study Analysis

Scenario:

A technology company has been contracted to develop a sophisticated facial recognition system intended for law enforcement applications. This system is designed to assist in identifying suspects by matching their facial features against a database. Initially celebrated for its high level of accuracy and efficiency in processing and identifying faces, the system, however, has come under scrutiny after reports emerged that it disproportionately misidentifies people of color. Investigations revealed that the misidentification rate for individuals of color was significantly higher than for white individuals, leading to a higher incidence of wrongful detentions and arrests among these communities. Further analysis indicated that the training data used to develop the algorithm lacked sufficient diversity, predominantly featuring faces of white individuals. This oversight has not only raised questions about racial bias inherent in AI technologies but also about the ethical implications of deploying such systems in sensitive areas like law enforcement without thorough vetting for fairness and impartiality.

1. What ethical problems are evident in this scenario?
2. What is one ethical approach to mitigate the bias found in the facial recognition system without compromising its accuracy?
3. How does the lack of diversity in training data contribute to the problem of algorithmic bias in this scenario?
4. What measures can be taken to ensure that the deployment of AI systems in law enforcement is both ethical and effective?
5. Discuss the potential long-term impacts on public trust and societal fairness if biases in AI systems like facial recognition are not addressed.

F. Ethical Dilemma

Read the following ethical dilemma and provide your response:

In a complex urban environment, an autonomous vehicle faces an imminent collision dilemma. The vehicle's path is obstructed by a pedestrian who has unexpectedly entered the roadway, violating traffic signals. Simultaneously, to the vehicle's side, a group of cyclists legally occupies the bike lane. The vehicle's AI must make an instantaneous decision: continue on its trajectory, risking harm to the pedestrian, or alter its course, endangering the cyclists. This decision-making process involves critical ethical considerations, including the valuation of human life, the assessment of potential harm, and the prioritization of legal and moral obligations in split-second scenarios.

Discussion Question: Consider the ethical dilemmas presented by AI technologies and explore how lawmakers, business leaders, and the public can work together to tackle these issues and encourage the ethical creation and implementation of AI.